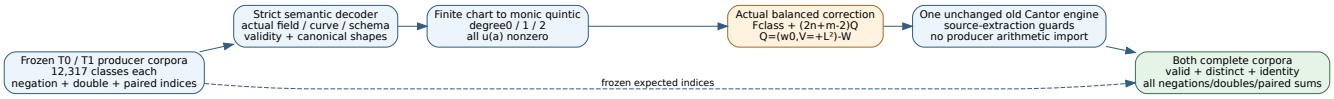
Independent arithmetic on both complete rational-class corpora

2026-11-12 · CT27 · exact T0/T1 models · scientific standing unsigned

The pre-existing generic Cantor engine now replays every one of the12,317 balanced classes of T0 and of T1. All source/quintic divisor validities, image distinctness, identity, negations, doubles and predetermined paired sums agree with frozen producer fixtures. Complete standalone03 passes 12tests, strictClippy and original source-extraction guards,PCC4b2aa9. The prior two49-class kernel replays remain. This is correctness evidence; no timing or isogeny-map membership result is claimed. [pcc:unverified]



The representation gap this closes

The earlier kernels had degree2,plus0 for every nonidentity class. Their ordinary-quintic baseline needed no additional infinity term. General balanced classes also have degree0/1 and every allowed infinity count. After the same branch chart, the correct ordinary class is the finite image plus(2n+m-2)Q. Only the degree0,plus1 input is the identity. Degree0,plus0/2 contribute-2Q/+2Q. These are exercised explicitly across both complete corpora. [pcc:unverified]

Each model	Degree0	Degree1	Degree2
T0	3	236	12,078
T1	3	236	12,078

Each model’s infinity correction counts for-2,-1,0,+1,+2 are [1,118,12079,118,1]. The actual identity index is239. These describe complete finite input coverage,not a cost or security estimate. Each class fixture is61,603bytes; each complete operator table is24,652bytes with an explicit model/schema header. No corpus is split to evade artifact limits. Producer records are dca7d7/6f943f. [pcc:unverified]

Existing implementation and custody

The checker imports only the existing flatFpR/polynomial operations and the old generic odd-model Cantor implementation from jv_cover.rs. The full original SHA256 is `1ecd62c4282402e3411635e0487d0bb36a5d57e15a547f352a7cab1fda7309cc`. All arithmetic bodies and exact extraction guards remain unchanged. Both replays share one engine module. No producer Balanced,tower, Richelot or elliptic-map code is imported. [pcc:unverified]

Why every infinity and finite support maps correctly

The same monic quintic chart

For a simple branch a, let L=F’(a),H(z)=z⁶F(a+1/z),w=L/(x-a), V=yw³/L and G(w)=L⁴H(w/L). G is a smooth monic quintic, with constant L⁴. The inverse is x=a+L/w,y=VL/w³. The unique quintic infinity W is the old branch point; the old infinities map to B±=(w0,V=±L²). Thus div(w)=B+ + B- −2W and Q-=-Q+ in Picard. The chart is over F101⁶; plain coefficient Frobenius is not asserted to fix its class images. [pcc:unverified]

For u=x²+b x+c,A=u(a), the support is U=w²+L(2a+b)w/A+L²/A. For u=x+u0 it is U=w+L/(a+u0). The ordinate is the remainder of v1w²+v(a)w³/L moduloU. These identities hold in the finite support algebra,including repeated jets. Every actual denominator and transformed curve equation is checked;no class is dropped. [pcc:unverified]

An input means Dfin+(n-1)infinity+ +(1-m-n)infinity-. Its transformed finite ordinary class is Fclass=[Dfin’-mW]. The principal-divisor relation gives the exact total Fclass+(2n+m-2)Q+. No torsion-free point-group or division-by2 assumption is used. The global isomorphism/Picard proof is FULL_CLASS_REPLAY_PROOF.md/PCCb71a87. Existing kernel-only40a568 stays immutable and keeps its earlier scope. [pcc:unverified]

Partial formalization and retained failures

Two individually bound BalancedChart statements db2184/9fd61b prove the new linear-support implication and the integer-module infinity accounting. Their local kernel/axiom gate03 passes2theorems,zero trusted native,with standard Lean foundations,aggregate7d9a53. Four existing KernelChart identities2026a6 cover the quadratic support and ordinate. Current source census134 excludes generated equations; actual fields,projective completion, valuations,principal divisors and Picard functoriality remain written. [pcc:unverified]

StrictClippy01 rejected duplicate module loading and fixed-size chunk parsing;diagnostics/source7b2f01 are retained. Final routing loads one unchanged engine and uses typed chunks. Formal scalar-normalization attempts01/02 failed;explicit module laws prove the identical statement03, efc7e5/affe42. No exemption,ignored test,accepted hole or history rewrite. [pcc:unverified]

Acceptance boundaries and continuation

The comparison covers every class,all negations/doubles and one fixed partner(37i+7)%12317 addition per class. It does not cover all151-million pairs or independently evaluate isogenies/kernel membership. Real consumer producer04/ef6945 reproduces all frozen bytes;complete changed guard05 passes25native/83reference unit+five integrations,one old ignored,all guards/fmt/strictClippy,93b147. Full101216-byte raw standard-XZ11984bytes has exact hashes/roundtrip;all source04/fixture04 hashes remain unchanged. Observation06 corrects only a clerical estimated timestamp in preserved05. Parent1498 exact25fd CI succeeds25native/82reference+five,full archive205514;this excludesCT27. New final-head CI/review and bottom-up integration remain separate. [pcc:unverified]

No measured stage,pipeline,index-calculus or rho advantage is established. Matched cold protocols remain owed after S3/livebudget admission. CT28 proposes direct F101 degree4 elliptic covers/norms to replace extension-field evaluation;no cover or gain is accepted. Larger fields/catalog,full geometric Lean,locked atlas and frozen-ledger integration remain open. [pcc:unverified]